

Assignment 8

$$1. \quad x(t) = \cos(\omega_0 t + \theta)$$

$$\theta \sim U[0, \pi]$$

$$f(\theta) = \frac{1}{\pi}$$

Ensemble avg $\rightarrow E[x(t)]$

$$\begin{aligned} E[x(t)] &= E[\cos(\omega_0 t + \theta)] \\ &= \frac{1}{\pi} \int_0^{\pi} \cos(\omega_0 t + \theta) d\theta \\ &= \frac{1}{\pi} [\sin(\omega_0 t + \theta)]_0^{\pi} \end{aligned}$$

$$= \frac{1}{\pi} [\sin(\pi + \omega_0 t) - \sin(\omega_0 t)]$$

$$= \frac{-2}{\pi} \sin(\omega_0 t)$$

Time avg $\rightarrow \frac{1}{2T} \int_{-T}^T x(t) dt$

$$\bar{x}_T = \frac{1}{2T} \int_{-T}^T \cos(\omega_0 t + \theta) dt$$

$$= \frac{1}{2T} \left[\frac{\sin(\omega_0 t + \theta)}{\omega_0} \right]_{-T}^T \rightarrow \frac{1}{2T} \left[\frac{\sin(\omega_0 T + \theta)}{\omega_0} - \frac{\sin(\theta - \omega_0 T)}{\omega_0} \right]$$

$$= \frac{1}{2T} \left[\frac{\sin(\omega_0 T + \theta) + \sin(\omega_0 T - \theta)}{\omega_0} \right]$$

$$= \frac{1}{2T} \frac{\sin(\omega_0 T) \cdot \cos(\theta)}{\omega_0}$$

$\therefore$ Time average $\neq$ Ensemble Average

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) dt = 0$$

$\therefore x(t)$ is not mean ergodic

$$2. \quad x(t) = 10 \cos(100t + \theta)$$

$$\theta \sim U[-\pi, \pi]$$

$$f(\theta) = \frac{1}{2\pi}$$

$$E[x(t)] = \frac{1}{2\pi} \int_{-\pi}^{\pi} 10 \cos(100t + \theta) d\theta$$

$$= \frac{5}{\pi} \left[\sin(100t + \theta) \right]_{-\pi}^{\pi}$$

$$= \frac{5}{\pi} \left[\sin(\pi + 100t) + \sin(\pi - 100t) \right]$$

$$= \frac{5}{\pi} \left[-\sin(100t) + \sin(100t) \right]$$

$$= 0$$

$$\bar{x}_T = \frac{1}{2T} \int_{-T}^T 10 \cos(100t + \theta) dt$$

$$= \frac{10}{2T} \left[\frac{\sin(100t + \theta)}{100} \right]_{-T}^T$$

$$= \frac{1}{20T} \left[\frac{\sin(100T + \theta) - \sin(-100T + \theta)}{100} \right]$$

$$= \frac{1}{20T} \left[\frac{\sin(100T + \theta) + \sin(100T - \theta)}{100} \right]$$

$$= \frac{1}{10 \cdot 20T} \left[\frac{2 \sin(100T) \cdot \cos(\theta)}{100} \right]$$

$$= \frac{1}{1000T} \left[\sin(100T) \cdot \cos(\theta) \right]$$

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) dt = 0$$

Ensemble average = Time average = 0

$\therefore$ Mean ergodic

$$3. \quad x(t) = 10 \cos(100t + \theta)$$

$$\theta \sim U[0, 2\pi]$$

$$f(\theta) = \frac{1}{2\pi}$$

$$E[x(t)] = E[10 \cos(100t + \theta)]$$

$$= \frac{1}{2\pi} \int_0^{2\pi} 10 \cos(100t + \theta) d\theta$$

$$= \frac{5}{\pi} \left[\sin(100t + \theta) \right]_0^{2\pi}$$

$$= \frac{5}{\pi} \left[\sin(2\pi + 100t) - \sin(100t) \right]$$

$$= \frac{5}{\pi} \left[\cancel{\sin(100t)} - \cancel{\sin(100t)} \right]$$

$$= 0$$

$$\bar{x}_T = \frac{1}{2T} \int_{-T}^T x(t) dt$$

$$= \frac{1}{2T} \int_{-T}^T 10 \cos(100t + \theta) dt$$

$$= \frac{5}{T} \left[\frac{\sin(100t + \theta)}{100} \right]_{-T}^T$$

$$= \frac{5}{T} \left[\frac{\sin(100T + \theta)}{100} + \frac{\sin(100T - \theta)}{100} \right]$$

$$= \frac{1}{1020T} \left[2 \sin(100t) \cdot \cos(\theta) \right]$$

$$= \frac{1}{10T} \sin(100t) \cdot \cos(\theta)$$

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) dt = 0$$

$$\therefore E[x(t)] = \bar{x}_T \rightarrow \therefore x(t) \text{ is mean ergodic}$$

$$4. \quad x(t) = A \cdot \cos(\omega_0 t + \theta)$$

$$\theta \sim U[-\pi, \pi]$$

$$f(\theta) = \frac{1}{2\pi}$$

$$E[x(t)] = E[A \cdot \cos(\omega_0 t + \theta)]$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} A \cos(\omega_0 t + \theta) d\theta$$

$$= \frac{A}{2\pi} \left[\sin(\omega_0 t + \theta) \right]_{-\pi}^{\pi}$$

$$= \frac{A}{2\pi} \left[\sin(\pi + \omega_0 t) + \sin(\pi - \omega_0 t) \right]$$

$$= \frac{A}{2\pi} \left[-\sin(\omega_0 t) + \sin(\omega_0 t) \right]$$

$$= 0$$

$$\bar{x}_T = \frac{1}{2T} \int_{-T}^T A \cos(\omega_0 t + \theta) dt$$

$$= \frac{A}{2T} \left[\sin(\omega_0 t + \theta) \right]_{-T}^T$$

$$= \frac{A}{2T} \left[\sin(\omega_0 T + \theta) + \sin(\omega_0 T - \theta) \right]$$

$$= \frac{A}{2T} \left[2 \sin(\omega_0 T) \cdot \cos(\theta) \right]$$

$$= \frac{A}{T} \sin(\omega_0 T) \cdot \cos(\theta)$$

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) dt = 0$$

$$\therefore E[x(t)] = \bar{x}_T$$

$\therefore$ Mean ergodic

$$5. \quad x(t) = \theta$$

$$\theta \sim U[-1, 1]$$

$$f(\theta) = \frac{1}{2}$$

$$E[x(t)] = E[\theta]$$
$$= \frac{1}{2} \int_{-1}^1 \theta \, d\theta$$

$$= \frac{1}{2} \left[\frac{\theta^2}{2} \right]_{-1}^1 \rightarrow \frac{1}{2} \left[\frac{1}{2} - \frac{1}{2} \right]$$

$$= 0$$

$$\bar{x}_T = \frac{1}{2T} \int_{-T}^T x(t) \, dt$$

$$= \frac{1}{2T} \int_{-T}^T \theta \, dt \rightarrow \frac{1}{2T} [\theta t]_{-T}^T \rightarrow \frac{1}{2T} [\theta T + \theta T]$$

$$= \frac{1}{2T} (2\theta T) = \theta$$

$$E[x(t)] \neq \bar{x}_T$$

$\therefore$ Not mean ergodic